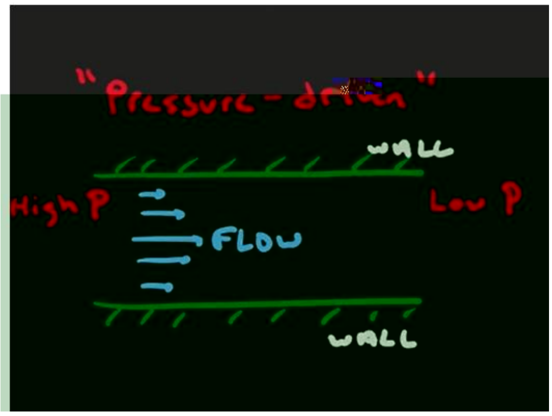
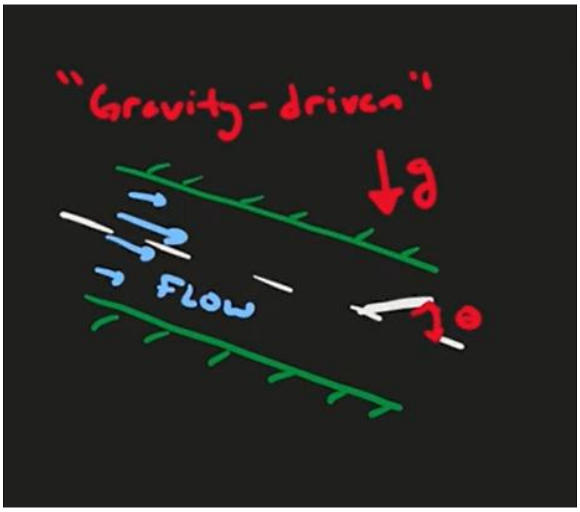
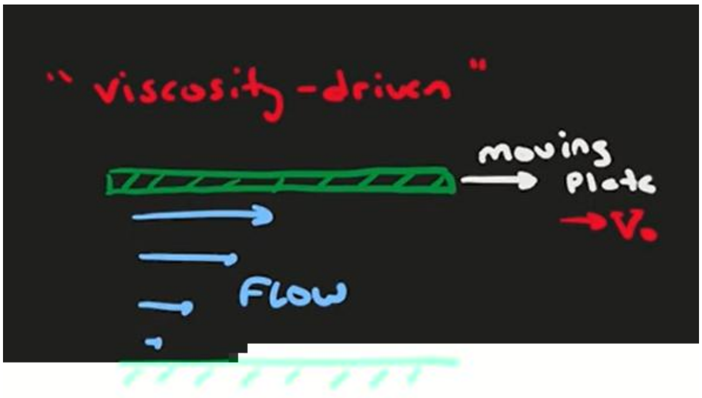
Aplicações simples das leis de conservação de massa e de conservação de quantidade de movimento - Escoamentos laminares internos fechados


"Pressure-driven"
WALL
High P
FLOW
Low P
WALL

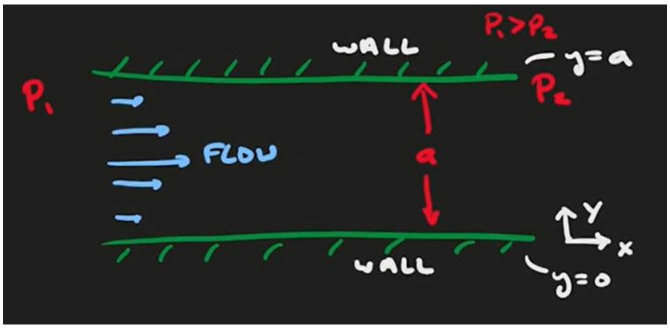




Escoamento plenamente desenvolvido dominado pelas forças de pressão

s s s s

d d



τ vw

$\frac{-}{w}$

$\frac{x}{}$

$\frac{-}{}$ z

τ

x 3 x d

Cz 3 z d

y 3 y d

$\frac{x}{}$ $C\frac{y}{}$ $\frac{z}{}$

$\frac{x}{}$ $\frac{z}{}$

$\frac{y}{}$

$-\frac{y}{} g \quad y \quad vw \quad F$

F

$x \quad _3 \qquad x \quad _d$

$x \quad _3 \quad -\frac{s}{} \qquad F \qquad F$

F

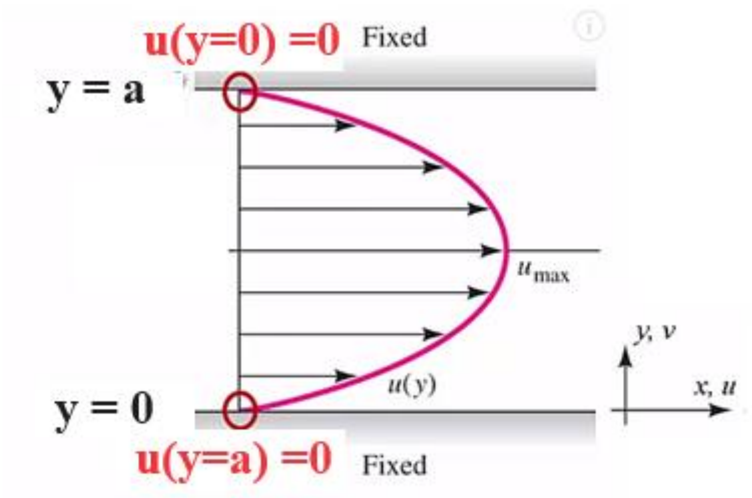
$x \quad _d \quad -\frac{s}{} d \qquad F \; d$

$F \qquad -\frac{s}{}\frac{d}{}$

$F \quad F$

$x \quad -\frac{s}{} \qquad -\frac{s}{}\frac{d}{}$

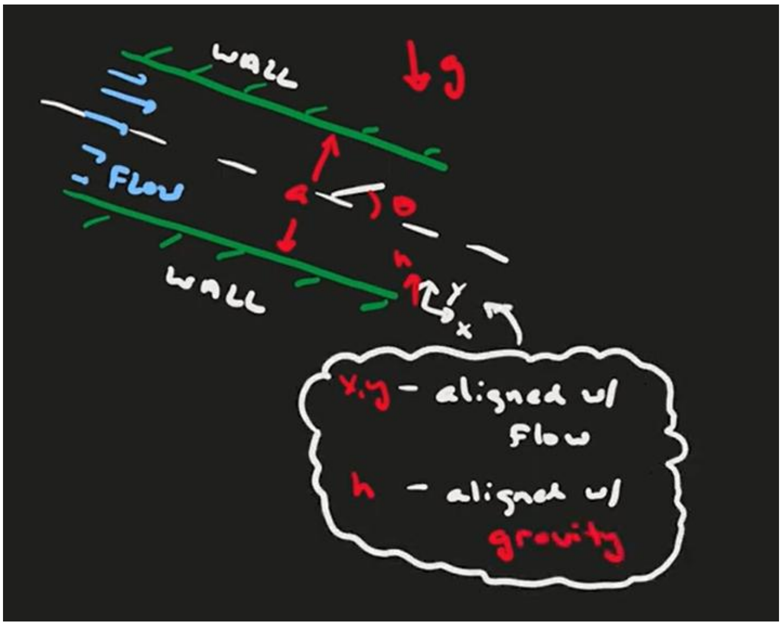
$x \quad -\frac{s}{} \qquad d$



Escoamento plenamente desenvolvido dominado pela força de corpo

d

d



τ vw

$\frac{-}{w}$

$\frac{x}{}$

$\frac{-}{}$ z

x 3 x d

Cz 3 z d

y 3 y d

$\frac{x}{}$ C $\frac{y}{}$ $\frac{z}{}$

$\frac{x}{}$ $\frac{z}{}$

$\frac{y}{}$

$\frac{y}{} g$ y vw F

y $_{3.d}$

F y

τ $\frac{x}{w}$ $x\frac{x}{}$ $y\frac{x}{}$ $z\frac{x}{}$ $\frac{s}{}$ $\frac{x}{}$ $\frac{x}{}$ $\frac{x}{}$ τ C

$\frac{x}{w}$

$x\frac{x}{}$ $x\frac{x}{}$

$z\frac{x}{}$ $\frac{x}{}$

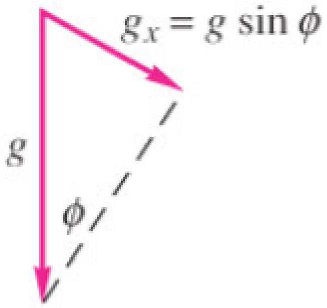
τ

y $y\frac{x}{}$

$\frac{s}{}$ $\frac{x}{}$ τ

$\frac{x}{}$ $\frac{s}{}$ τ

vlq



vlq $\frac{g}{g}$

$\frac{x}{} \quad -\frac{s \quad \tau}{} \quad vw$

$\frac{x}{} \quad -\frac{s \quad \tau}{} \quad F$

x

$x \quad -\frac{s \quad \tau}{} \quad F \quad F$

$x \quad 3 \quad x \quad d$

$x \quad 3 \quad -\frac{s \quad \tau j}{} \quad F \quad F$

F

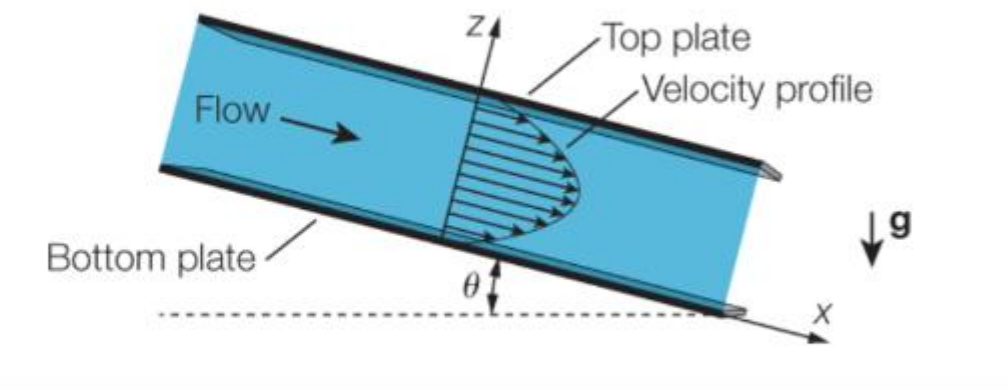
x d $- \frac{s \ \tau j}{} \frac{d}{}$ $F \ d$

F $- \frac{s \ \tau j}{} \frac{d}{}$

F F

x $\frac{}{} \frac{s \quad \tau}{}$ $- \frac{s \quad \tau}{} \frac{d}{}$

x $\frac{}{} \frac{s \quad \tau}{}$ d



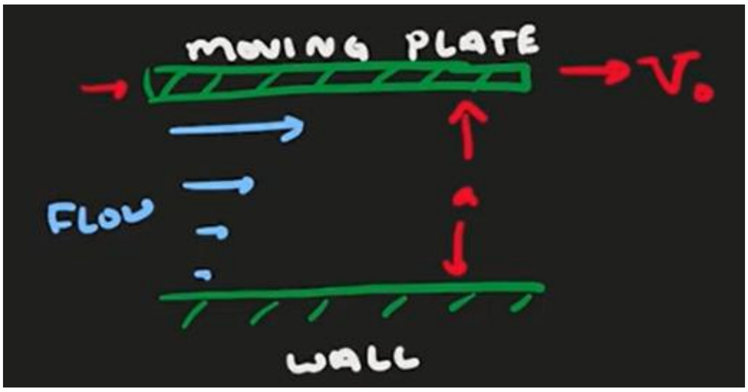
Escoamento plenamente desenvolvido dominado pelas forças friccionais

V s

s s s

d

d



τ vw

$\frac{-}{w}$

$\frac{x}{}$

$\frac{-}{}$ z

x $_3$ x $_d$ $_3$

$\mathrm{C}z$ $_3$ z $_d$

y $_3$ y $_d$

$\frac{x}{}$ $\mathrm{C}\frac{y}{}$ $\frac{z}{}$

$\frac{x}{}$ $\frac{z}{}$

$\frac{y}{}$

$\frac{y}{}g$ y vw F

y $3.d$

F y

τ $\frac{x}{w}$ $x\frac{x}{}$ $y\frac{x}{}$ $z\frac{x}{}$ $\frac{s}{}$ $\frac{x}{}$ $\frac{x}{}$ $\frac{x}{}$ τ C

$\frac{x}{w}$

$x\frac{x}{}$ $x\frac{x}{}$

$z\frac{x}{}$ $\frac{x}{}$

τ

y $y\frac{x}{}$

$\frac{x}{}$

$\frac{x}{}$ F

x

x F F

x 3 x d 3

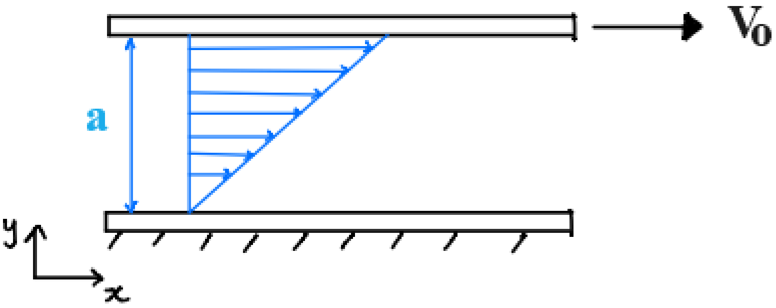
x 3 F F

F

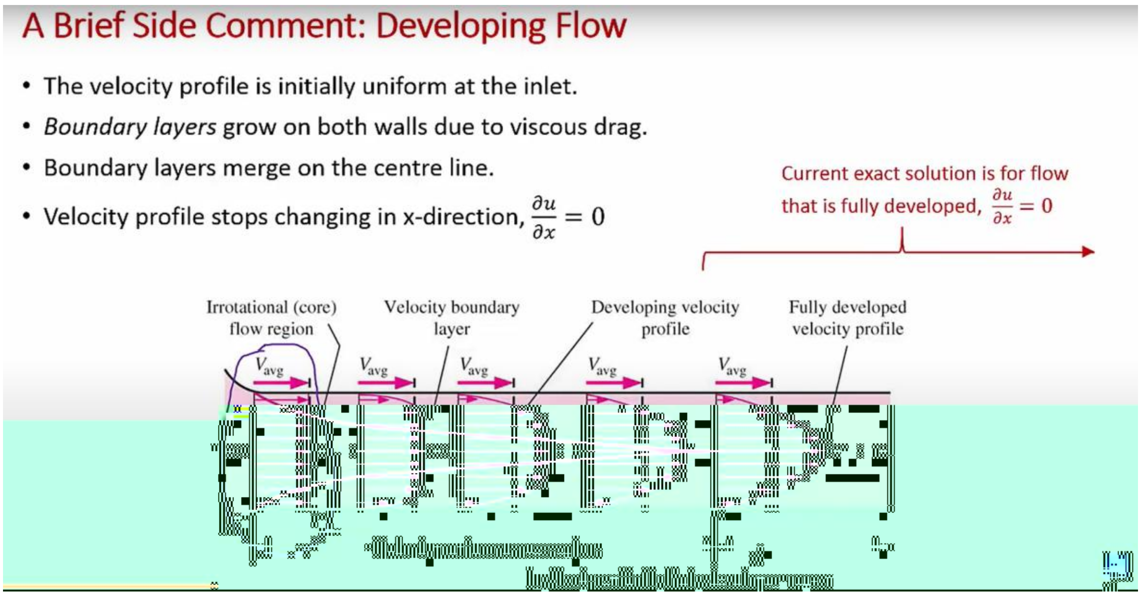
x d F d 3

F $\frac{}{d}$

x $\frac{3}{d}$ d



V_0
a
y
x



A Brief Side Comment: Developing Flow
• The velocity profile is initially uniform at the inlet.
• *Boundary layers* grow on both walls due to viscous drag.
• Boundary layers merge on the centre line.
• Velocity profile stops changing in x-direction, $\frac{\partial u}{\partial x} = 0$
Current exact solution is for flow that is fully developed, $\frac{\partial u}{\partial x} = 0$
Irrotational (core) flow region
Velocity boundary layer
Developing velocity profile
Fully developed velocity profile
V_{avg}